

DAVID MCDUFFIE

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Summary

Results-driven **AI professional** with **9+** years of hands-on experience delivering scalable, production-ready solutions across **LLMs**, recommendation engines, and full-stack systems. Proven expertise in building and orchestrating **LLM-powered AI agents** using cutting-edge tools like **LangGraph**, **LangSmith**, and **PyTorch**, coupled with deep knowledge of **e-commerce personalization**, **multimodal NLP**, and **MLOps** pipelines.

- ❖ **Led LLM agent development** with memory, API access, and dynamic branching logic, elevating user engagement and automation.
- ❖ Delivered **23% boost in conversions** and **37% engagement lift** through **hyper-personalized product recommendations** powered by **user embeddings**.
- ❖ Architected scalable **AI platforms** with robust **observability**, **CI/CD** pipelines, and real-time analytics using **cloud-native tools (AWS, SageMaker, EMR)**.
- ❖ Drove multilingual **NLP** applications at Meta, scaling to **1B+ users** with low-latency, distributed inference.
- ❖ Implemented **mission-critical systems** including **fraud detection**, **secure authentication**, and **AI content moderation**, improving **KPIs** and compliance.
- ❖ Trusted mentor and cross-functional leader, collaborating with product, **DevOps**, and design teams to deliver impactful **AI-powered** user experiences.

Combining deep technical fluency with strategic vision, I excel in driving innovation from ideation to execution across full-stack AI systems.

Experience

Senior AI Full Stack Engineer

11/2023 - Current

Simform | San Francisco, CA | Remote

- Spearheaded the development of **LLM-powered AI agents** using **LangGraph** and **LangSmith**, enabling dynamic multi-turn conversations and tool-chaining capabilities for intelligent automation.
- Built and deployed a **hyper-personalized product recommendation system** for e-commerce, leveraging **LLMs** and user **embeddings** to drive a **23% increase in conversions** and **37% boost in engagement**.
- Designed and implemented a context-aware orchestration layer for **AI agents** with **LangGraph**, supporting **memory**, external API access, and decision branches for personalized user experiences.
- Utilized **LangSmith** for observability, experiment tracking, and prompt evaluation, establishing continuous improvement pipelines for **AI agent** performance tuning.
- Engineered **user behavior analytics pipelines** with **embeddings** and real-time signals to enhance product **recommendations** and **personalize** the shopping journey at scale.
- Conducted extensive **A/B testing** and performance tuning for recommendation algorithms, monitoring user engagement and iteratively refining the **AI models** based on feedback and key performance indicators (**KPIs**).
- Partnered with **product**, **design**, and **DevOps teams** to ensure reliable AI feature delivery, optimized for low-latency inference, high uptime, and robust monitoring.
- **Mentored** junior developers and engineers, leading **code reviews**, implementing best practices for coding standards, and providing technical guidance on integrating AI solutions into the overall software architecture for continuous improvement and innovation.

AI Engineer

01/2021 - 09/2023

Meta AI | Menlo Park, CA | Remote

- Led the end-to-end integration of **recommendation engine** models into **Facebook** Feed and **Instagram** Explore, boosting user engagement metrics (**CTR**, dwell time) by **over 12%** through contextual content ranking.
- Designed and deployed large-scale **NLP models** for **Messenger** to enable smart replies, intent recognition, and multi-turn conversations, handling **over 500 million daily interactions** with low-latency response times.
- Collaborated with product teams to implement **vision-language transformers** for **Instagram Stories** and **Reels**, enabling real-time **content moderation** and automatic captioning, improving accessibility compliance by 30%.
- Developed and optimized **personalized AI notification systems** using **reinforcement learning** across **Facebook** and **Instagram**, improving push notification **click-through rates by 18%** while reducing user churn.
- Contributed to the deployment pipeline of multilingual **LLM-based chat assistants** within Messenger, enabling seamless in-app customer support in **15+ languages** with human-in-the-loop training workflows.
- Utilized Meta's internal **PyTorch/XLM** infrastructure to scale AI inference across 1B+ users, implementing distributed model serving and A/B testing frameworks for fast iteration and performance tracking.

Machine Learning Engineer

05/2018 - 12/2020

Constructor | San Francisco, CA

- Developed and deployed ML models for AI-powered search ranking and autocomplete optimization, improving average user conversion rate by 6.5% across multiple enterprise clients.
- Designed personalized product recommendation engines using clickstream behavior, session embeddings, and collaborative filtering, driving measurable uplifts in CTR and engagement metrics.
- Built real-time inference pipelines for query intent classification using NLP and Transformer-based models, enhancing autocomplete suggestion accuracy and reducing bounce rate by 18%.
- Optimized A/B testing infrastructure for ranking models and recommendation widgets, integrating Bayesian Optimization and multi-armed bandit strategies to accelerate experiment velocity.
- Engineered and maintained a centralized feature store capturing session, product, and customer context, reducing data duplication and model onboarding time by 30%.
- Designed anomaly detection modules for identifying low-quality or manipulated query inputs and protecting the recommendation engine from noisy event data.
- Created end-to-end ML pipelines on AWS (Lambda, EMR, Glue), integrating with CI/CD via GitHub Actions and Docker-based containers for scalable deployments.
- Strengthened platform security by implementing OAuth 2.0, JWT-based API access, and MFA, resulting in a 75% drop in unauthorized access attempts across customer-facing endpoints.
- Mentored junior engineers on MLOps best practices and model explainability techniques, increasing team-wide productivity and improving debugging efficiency for live model issues.

Backend Engineer

06/2016 - 04/2018

Zappos | Las Vegas, NV

- Designed and implemented scalable **microservices** architecture to support the e-commerce platform's core functionalities, improving backend performance by 30% and enabling smoother integration with frontend systems and third-party APIs.
- Optimized existing **RESTful APIs** and database queries, reducing response times by over 40% for high-traffic endpoints. Introduced asynchronous processing using message queues (e.g.,

- **RabbitMQ** or **Kafka**) to handle order processing and user notification systems with high reliability.
- Led the migration from monolithic systems to **cloud-native infrastructure (AWS)** by containerizing legacy services with Docker and orchestrating deployments using **Kubernetes**.
- Collaborated cross-functionally with product managers, frontend developers, and QA teams to deliver new features such as personalized product recommendations and real-time inventory management, while maintaining high code quality and automated test coverage above 90%.

Software Developer Intern

07/2015 - 05/2016

Zappos | Las Vegas, NV

- Collaborated with senior backend engineers to design and implement scalable **RESTful APIs** supporting customer personalization and order tracking, reducing response **latency** by 20%.
- Built and optimized database queries in **MySQL** and **DynamoDB**, improving data retrieval efficiency for high-traffic services on the Zappos.com platform.
- Participated in **Agile development** cycles, contributing to code reviews, **unit test** coverage, and CI/CD deployment pipelines using **Jenkins** and **Docker**.

Education

Bachelor of Science: Computer Science

05/2016

University of Nevada, Las Vegas | Las Vegas, NV

Skills

- **Artificial Intelligence & Machine Learning**
LangGraph, LangSmith, OpenAI API, Hugging Face Transformers, PyTorch, TensorFlow, Scikit-learn, XGBoost, Reinforcement Learning, Smart Replies, Multi-turn Dialogue Systems, Prompt Engineering, Memory-Augmented Agents, Multilingual NLP, Vision-Language Models, Hyper-personalized Recommendations, User Embeddings, CTR Optimization, Behavioral Analytics, A/B Testing, Bayesian Optimization, Feature Stores, Model Retraining, Human-in-the-loop Training, Experiment Tracking
- **Backend Development**
Python, Java, C++, JavaScript (Node.js), SQL, RESTful APIs, Microservices Architecture, Asynchronous Processing, Event-driven Systems, Distributed Systems, Kafka, RabbitMQ, MySQL, DynamoDB, Redis, NoSQL
- **Cloud & Infrastructure**
AWS (Lambda, EC2, S3, SageMaker, EMR, Glue), Docker, Kubernetes, Jenkins, GitHub Actions, AWS CodePipeline, GitLab CI/CD
- **Security & Performance**
OAuth 2.0, JWT, Multi-Factor Authentication (MFA), Rate Limiting, Secure API Access, Low-latency Optimization, Performance Monitoring, Observability, Monitoring Dashboards
- **Collaboration & Leadership**
Agile/Scrum, Cross-functional Team Collaboration, Technical Mentoring, Code Reviews, Product Integration, Knowledge Sharing, Documentation, Scalable Architecture Planning